Post-Quantum Cryptography Migration

Part 6 - Protocols & Trust: Network & Transport Protocols

Compiled by the Spaced Research Ledger

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The protocols that move data across networks are where post-quantum cryptography first became real. Every encrypted connection made today can be recorded and decrypted later by a quantum computer, so key exchange had to move first, years ahead of signatures.

This report covers seven protocol families. Section 1 covers TLS, the web's transport security, and Section 2 its younger siblings QUIC and DTLS. Section 3 covers IPsec and the VPNs, Section 4 SSH, and Section 5 the link and access layer inside networks. Section 6 covers DNS and time, and Section 7 the routing and storage protocols.

The milestone of 2026 is that hybrid key exchange crossed from experiment to majority. Post-quantum key exchange now covers over half of the web requests one large network observes. The defining RFC was published in August, and Akamai runs it end to end from browser to origin. SSH is on the same path, with the newest OpenSSH warning users when a connection is not quantum-safe.

The unsolved problems are just as clear. Certificates and signatures everywhere remain classical. DNSSEC has no good answer yet for signatures too large for its packets. And the further from the web's center a protocol sits, the less has happened, with storage protocols like iSCSI showing no migration activity at all.

1. TLS 1.3 Hybrid Key Exchange

TLS is the protocol behind the padlock in a browser, securing most traffic on the internet. Hybrid key exchange runs a classical algorithm and ML-KEM together, so a session is safe unless both break. It defends against the one quantum threat that applies today: recording encrypted traffic now to decrypt later. That is why it leads the whole migration.

Recent Developments

The standards finally caught up with the deployments. RFC 10024, published August 2026, defines the three hybrid groups as a Proposed Standard and formally retires the pre-standard draft groups [1]. RFC 9954, published July, provides the general hybrid framework beneath them [2]. Akamai reached general availability for end-to-end post-quantum connections across its network in August, covering every leg from client to origin [3]. Java's hybrid key exchange feature was delivered for JDK 27 [4], placed first in the default group list [5].

Adoption crossed the halfway mark. Post-quantum key exchange rose from about 29 percent of one network's HTTPS requests in mid-2025 to 54 percent in mid-2026, far faster than other protocol transitions [6]. A domain-level study puts partial readiness at 49 percent of domains, a reminder that request-weighted figures lean on high-traffic sites [7]. Library and server support is now broad across mainstream stacks [7].

Defaults keep flipping on. OpenSSL negotiates the hybrid group with no application changes [8], and Windows and ASP.NET reached general availability in July [9]. Google Cloud's load balancers are on a staged rollout ending with no opt-out in 2027 [10], and even a pure-Erlang QUIC stack added the group [11]. Go's default group list now carries all three hybrids [12].

Current State

One group carries nearly all the traffic. X25519MLKEM768 is the default for Chrome, Firefox, and Edge, and accounts for roughly 95 percent of the post-quantum TLS one large network observes [13]. The IETF adopted the defining draft in March 2025 with three groups [13].

The P-256 variant exists specifically for FIPS environments, since X25519 is not FIPS-approved [14]. Hybrid key exchange is the most widely deployed post-quantum cryptography on the internet precisely because harvest-now-decrypt-later applies immediately [14]. The old draft group is formally superseded, with guidance to retire it [14], though F5's BIG-IP still spoke only the incompatible draft as of one mid-2025 assessment [15].

Two operational wrinkles persist. Go lists the X25519 hybrid as FIPS-approved but rejects its classical half under the strictest FIPS mode, breaking the handshake and forcing the P-256 variant instead [13]. And Akamai had already defaulted the hybrid group for customer origin connections back in January [16].

Unresolved Conflicts

Apple's timeline is the one genuine muddle. Sources disagree on when Safari and Apple's operating systems turned hybrid TLS on by default [17]. One dates it to 2025 and another to the 26-generation systems in 2026, against a backdrop where Apple's messaging protocol got more publicity than its TLS work [17]. A separate source asserts Safari 18 ships the hybrid on by default, adding a data point without resolving the inconsistency [18]. An Apple release note naming the version would settle it.

2. QUIC, HTTP/3, DTLS

QUIC is the transport under HTTP/3, carrying a growing share of web traffic over UDP instead of TCP. DTLS secures datagram traffic like video calls and VPN tunnels. Both borrow the TLS 1.3 handshake, so they inherit post-quantum key exchange rather than inventing their own. The interesting questions are about packet sizes, not cryptography.

Recent Developments

The specifications made inheritance official. RFC 9954 instructs that hybrid groups registered under its framework should be marked usable in DTLS [19], and RFC 10024 registers all three groups accordingly [20]. The practical caveats are dimensional: DTLS handshakes fragment more once post-quantum keys enlarge them [21], and QUIC's 1,200-byte initial packets force extra packets in the first flight [21]. Firefox supports the hybrid group on QUIC connections from version 135 [22]. At the experimental edge, a Rust crate now implements pure post-quantum QUIC with no classical fallback at all [23].

Current State

No QUIC-specific protocol change was ever needed, because QUIC carries TLS 1.3 as its handshake [24]. OpenSSL lists QUIC server support within the same post-quantum feature set as its default hybrid group, with zero-round-trip resumption still in development for QUIC specifically [25]. DTLS compatibility was a design goal from the earliest drafts of the hybrid groups, not a late addition [26]. The performance story even favors the newer transport: a comparative study found QUIC reduces the latency and bandwidth penalty of post-quantum handshakes relative to TLS, especially on lossy networks [27].

3. IPsec & VPN Protocols

IPsec secures site-to-site links and corporate VPNs, negotiated by a protocol called IKEv2. The consumer VPN world mostly runs on WireGuard or OpenVPN instead. All three took different post-quantum paths: IKEv2 grew standardized extensions, OpenVPN inherits from OpenSSL, and WireGuard, which refuses algorithm negotiation on principle, gets its protection bolted on from outside.

Recent Developments

The IKEv2 standards pipeline is producing steadily. The draft specifying ML-KEM in IKEv2 reached its ninth revision [28]. A downgrade-prevention mechanism for hybrid setups cleared review and entered the RFC Editor queue [29]. Further drafts cover variable-length PRFs [30], reliable transport for hybrids [31], and FrodoKEM [32].

strongSwan’s experimental branch now runs fully post-quantum tunnels, pairing ML-DSA certificates with ML-KEM key exchange [33]. Palo Alto’s implementation supports an unusually broad KEM menu, from ML-KEM through Classic McEliece [34].

OpenVPN’s ecosystem matured through its library. Recent releases build against newer OpenSSL versions carrying the algorithms [35], with a security patch release following [36]. A production VPN provider demonstrated fully post-quantum OpenVPN with ML-DSA authentication [37], and the Easy-RSA certificate tooling gained first-class post-quantum support [38]. On the WireGuard side, NetBird embedded the Rosenpass key exchange for automatic quantum-safe key rotation on every connection [39].

Current State

IKEv2 itself was never revised. Everything rides on two extensions. RFC 8784 mixes an out-of-band preshared key into the exchange, keeping session keys safe from a quantum adversary if the secret has enough entropy [40]. It transmits only a key identifier in-band [41] and mixes rather than replaces the classical exchange [42].

Formal verification confirms the secrecy claim while noting some authentication properties are not restored [43]. Vendors treat it as the deployable near-term mitigation [44]. RFC 9867 extended the mechanism to later exchanges for full protection and fresh rekeying [45].

RFC 9370 is the real hybrid framework. It allows up to seven additional key exchanges of any kind [46]. The large payloads travel in an intermediate exchange [46], and classical and post-quantum methods merge into one negotiation registry [46]. It deliberately defines no concrete combinations, leaving those to separate drafts [47].

In practice the design shows on the wire: the small classical key goes first, the large ML-KEM key follows in the intermediate exchange, fragmented to fit protocol limits [48]. strongSwan shipped it in version 6.0.0 [49]. Palo Alto trades its agility against fragmentation and a larger pre-authentication attack surface [34]. Cloudflare runs hybrid post-quantum IPsec in production [50], upgraded across its appliance line [51].

OpenVPN inherits everything from OpenSSL, so capability arrives with the host system's library version [52]. Its Connect client still lacks support, promised for a future release [52]. The full algorithm set works where OpenSSL 3.5 is present [53], and its handshake-encrypting preshared-key feature has quietly protected against harvest-now-decrypt-later for years [53]. NordVPN's post-quantum feature explicitly excludes OpenVPN [54].

WireGuard's story is deliberate inflexibility. The protocol refuses algorithm negotiation, so post-quantum resistance comes from outside mechanisms [55], canonically by mixing a separately established post-quantum key into its preshared-key slot [55].

Rosenpass does exactly that, refreshing the key every couple of minutes [56]. Mullvad enabled the same pattern by default across desktops in January 2025 [57] and offers a standalone tool [58]. Surfshark followed in January 2026 [56], NordVPN integrated it into its WireGuard-based protocol with listed exclusions [54], and ProtonVPN abandoned WireGuard compatibility entirely for a ground-up redesign [59]. Academic proposals that rebuild the handshake itself remain research, hampered by very large keys [60].

4. SSH

SSH is how administrators log into servers and how code moves between machines. Its post-quantum story is further along than almost any other protocol, because one implementation dominates and its maintainers moved early. OpenSSH has defaulted to post-quantum key agreement since 2022, before NIST had even finished standardizing.

Recent Developments

OpenSSH started on signatures. Version 10.4 added experimental support for a composite post-quantum host-key scheme combining ML-DSA-44 with Ed25519, disabled by default [61]. Version 10.5 followed with fixes and a performance improvement for the older post-quantum key agreement [61], which Debian promptly packaged [62].

Server-side scans show post-quantum key exchange on about 11.8 percent of SSH servers, nearly double the prior year [63]. One short-lived hybrid was added and removed within a month [64]. AWS's file-transfer service added the full set of ML-KEM hybrids to its security policies [65], and even an iOS client app now negotiates them [66].

Current State

OpenSSH's defaults have led the industry for years. Post-quantum key agreement has been the default since version 9.0 in April 2022, first with a pre-NIST lattice scheme, then with the ML-KEM hybrid becoming default in version 10.0 [67]. Since 10.1 the client actively warns when a connection falls back to classical key agreement, naming the store-now-decrypt-later risk, with a config option to silence it [67].

The ML-KEM hybrid itself arrived in 9.9 [68], is built so both components must break [67], and its method names are being registered through an IETF draft [69]. The older scheme was standardized retroactively as RFC 9941 in April 2026 [70], having been created in 2020 [71] and made default in 2022 [71]. Version 9.9 gave it a standardized identifier while keeping its legacy vendor name [72].

Deployment follows distribution schedules. Current Debian, Ubuntu, and RHEL releases ship post-quantum-capable OpenSSH, while the older Ubuntu LTS needs a backport [73]. Ubuntu 26.04 makes the ML-KEM hybrid the default [74]. The installed base lags: roughly three-quarters of observed servers run versions predating default post-quantum support [75].

GitHub turned it on for its SSH endpoints in fall 2025 [73], starting outside the United States [76]. SUSE enabled the older hybrid through its crypto policies [77], and RHEL adds FIPS-compatible hybrids through downstream patches, with its test policy enabling only ML-KEM by default [78]. The pre-NIST scheme was never a NIST algorithm at all [78].

The alternative implementations kept pace. PuTTY 0.83 supports ML-KEM, hybrid-only by design [79]. Dropbear added both post-quantum exchanges in 2025 [80], through community contributions, and recommends enabling at least one despite the code-size cost [80]. libssh added and prefers the ML-KEM hybrid [81], Termius [82] and Tectia followed, with Tectia removing an obsolete scheme [83], and AWS's SFTP service interoperates with the post-quantum clients [84][85].

5. Link & Access Layer

Below the internet protocols sit the technologies that secure individual network links: MACsec encrypting traffic between switches, and 802.1X deciding who may join a network at all. Their bulk encryption is symmetric and already quantum-resistant at 256 bits. The soft spot is the same as everywhere else: the key exchange and certificates that set the sessions up.

Recent Developments

The vendors committed to dates. Cisco pledged post-quantum integration across MACsec, IPsec, TLS, and SSH for the majority of its core portfolio by December 2026 [86]. RAD demonstrated a quantum-safe 400-gigabit MACsec system using hybrid ML-KEM [87]. On the research side, a hybrid key exchange protocol integrates ML-KEM, ML-DSA, and QKD into MACsec without touching the IEEE standards [88]. RADIUS, by contrast, shows no protocol-specific migration activity at all, with available material addressing only the TLS layer beneath it [89].

Current State

MACsec's data plane was never the problem. The standard's 256-bit AES modes are considered quantum-resilient [90], added by amendments over a decade ago [90], and guidance for quantum-secure configuration centers on 256-bit keys and functions throughout [91]. The companion MKA protocol establishes those keys, over either preshared keys or 802.1X authentication [91][91]. Cisco's own guidance is candid that the EAP-TLS path cannot be called quantum-secure until TLS itself is [91].

The real fixes are arriving: ML-KEM-based MACsec key exchange in IOS-XE 26.1 [92], and a dedicated configuration guide for hybrid ML-KEM with EAP-TLS on its router lines [93]. A full-stack post-quantum claim for its newest switches is best read against the layered maturity beneath it [94].

A QKD-integrated key-delivery mode exists on data-center switches [95], and a lab demonstration hit 196 gigabits of quantum-secure MACsec on commodity offloads [96]. FIPS mode, for now, excludes the hybrid algorithms [97]. The research protocol detailing hybrid MACsec keying published in 2025 [98].

Access control inherits TLS's schedule. EAP-TLS gets post-quantum security from the TLS handshake beneath it [99], and its certificate-based deployments will eventually need post-quantum or hybrid certificates [100]. Today's rollout covers key agreement while certificate chains stay classical [101].

The IETF work has been adopted onto the standards track [102], building on earlier drafts addressing large certificates in constrained EAP exchanges [103]. Cisco's network-access product extended TLS 1.3 to its EAP methods as a prerequisite step [104]. Measured deployments look workable, with session resumption absorbing most of the added latency from larger handshakes [105]. MACsec's symmetric suites themselves need no negotiation change for the quantum era [106].

6. DNS & Time

DNSSEC signs the internet's address book so answers cannot be forged, and encrypted DNS carries queries over TLS. Time synchronization has its own security protocol. DNS is the migration's hardest packaging problem: post-quantum signatures simply do not fit in the small UDP packets DNS was built on, and no standard answer has been chosen.

Recent Developments

The DNSSEC proposals are multiplying because none has won. One draft would let resolvers apply local algorithm policy so quantum-safe signatures can ride alongside legacy ones [107]. Another splits algorithms by role, big conservative ones for key-signing keys and compact ones for zone signing [108]. A third applies Merkle Tree Ladder mode to ML-DSA to amortize its signature size [109].

Fresh measurements at country-level domain operators found the MAYO-2 candidate outperforming classical RSA signing, while Falcon-512 matches it with larger signatures [110]. An ACM paper argues for exactly this kind of differentiated approach [111]. The TLS-application guidance now explicitly covers encrypted DNS [112], and a public test tool checks post-quantum support on the DNS-over-TLS port [113].

Current State

The size arithmetic frames everything. DNS mostly runs over UDP with about 1,232 usable bytes, while ML-DSA signatures run 2,420 bytes and beyond, and SLH-DSA reaches nearly 8,000 [114][115]. Every post-quantum family tested pushes responses past UDP limits and into TCP fallback [116], with the smallest responses coming from candidates still under evaluation [116].

The strategy taking shape is dual-algorithm: a conservative scheme in an impact-mitigating mode, paired with a compact drop-in [115]. Merkle Tree Ladder mode amortizes signatures across responses [115], with named candidates in each role [115]. Prototypes exist, including a CoreDNS plugin wiring thirteen post-quantum algorithms through liboqs [117], and a protocol proposal to cut the TCP fallback cost [118]. U.S. government guidance already lists DNS software among the product categories expected to adopt the standards [119].

Encrypted DNS moves with TLS. AdGuard enabled hybrid key exchange by default across its DNS transports, describing itself as an early adopter just behind Google and Cloudflare [120]. Chrome's ML-KEM rollout covers its DNS-over-HTTPS connections [121], Cloudflare's origin infrastructure includes its DNS service [121], and ICANN's technical office has assessed the implications across the encrypted transports [122].

AWS's endpoint migration quantifies the cost, about 1,600 extra handshake bytes and well under a millisecond of compute [123]. The hybrid TLS draft applies directly to DNS-over-TLS [124], whose negotiated key share grows to 1,216 bytes [124], on a rollout underway since 2024 [125]. A post-quantum DNS-over-TLS framework in the literature remains simulation-only [126].

Time is the quiet corner. Network Time Security, standardized in 2020, splits key establishment over TLS from the lightweight synchronization protocol [127], and remains offered by few services [128]. No post-quantum-specific NTS standardization work was found at all [127].

7. Routing & Storage Protocols

The protocols that steer internet routes and move stored data sit furthest from the post-quantum frontier. RPKI, which vouches for who may announce which internet addresses, has active design debates. The storage protocols mostly rely on symmetric encryption or have simply not started. This section is short because the activity is.

Recent Developments

RPKI's design argument is now on paper. One draft reframed itself from a production algorithm profile into an experiment report on post-quantum signatures and migration approaches [129]. A competing proposal adds an additive post-quantum layer, protecting the repository root with a Merkle tree ladder while leaving existing objects untouched and legacy-compatible [130]. A related draft covers router certificates while explicitly not changing the BGPsec message signature algorithm [131]. For iSCSI, no post-quantum migration activity was found at all [132].

Current State

RPKI's open question is how much to preserve. Backward compatibility and a long dual-stack transition complicate adoption, as in other PKI ecosystems [133]. The working group has pushed back on stripping objects to save signature bytes, since that breaks existing structures [133]. The Merkle-tree approach amortizes one post-quantum signature across many objects [133] and stays compatible with existing size-reduction techniques [133]. No ratified standard exists, and the integrity debate continues [133].

The file protocols are symmetric stories. SMB 3.1.1 gained 256-bit AES modes on current Windows [134] and negotiates its cipher at connection time [135]. It defaults to the 128-bit mode for performance [134], extends encryption to its RDMA transport [134], and is mirrored by Azure Files [136]. Nothing post-quantum is specified in SMB's negotiation at all [135].

NFS security rides Kerberos through the RPCSEC_GSS mechanism [137], a multiplexing layer dating to 1999 [138]. It is split between kernel and userspace on Linux [139], version-limited on mainframes [140], and unevenly implemented historically [141]. Its weakest common mode still authenticates by user ID alone, with no cryptography to migrate [141].

About This Report

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